

Thiago E. Kalid

MASTER'S STUDENT AT LASSIP

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Education

UTFPR (Federal University of Technology – Paraná)

Curitiba, Brazil

M.Sc. IN COMPUTER SCIENCE AND ELECTRICAL ENGINEERING

Mar. 2025 - Present

- Advisor: Thiago A. R. Passarin
- Research topics: Signal Processing, Ultrasonic imaging, Beamforming, Phased-Array, Machine Learning, Inverse-problems.

UP (University of Porto)

Porto, Portugal

UNDERGRADUATE EXCHANGE STUDENT

Aug. 2023 - Feb. 2024

- Completed classes: Audio Signal Processing, Machine Learning, Forecasting.

UTFPR (Federal University of Technology – Paraná)

Curitiba, Brazil

B.Sc. IN ELECTRICAL ENGINEERING (*Summa Cum Laude*)

Mar. 2019 - Mar. 2025

- Thesis title: Development of a Visual Odometry Method for Ultrasonic Immersion Inspection
- Graduated 1st in class.

Research Experience

LASSIP - Laboratory of Statistical Signal Processing & Inverse Problems

Curitiba, Brazil

GRADUATE RESEARCH ASSISTANT

Mar. 2025 - Present

- **Advisor:** *Thiago A. R. Passarin*
- Developed a Python ultrasound simulator, based on ray-tracing, to support and optimize acoustic lens design.
- Designed and implemented a machine learning-based (and a deep learning-based) anomaly detection framework for flaw detection in ultrasonic non-destructive testing of pipes.
- Developed a framework for refraction and reflection corrected beamforming through multiple media.
- Designed an acoustic lens capable of statically inspecting large pipe sections (e.g., 90°), significantly reducing inspection time in deepsea inspections.

LASSIP - Laboratory of Statistical Signal Processing & Inverse Problems

Curitiba, Brazil

UNDERGRADUATE RESEARCHER

Apr. 2021 - Aug. 2023

- **Advisor:** *Thiago A. R. Passarin*
- Developed an image-processing-based system for underwater 2-D position estimation using a Raspberry Pi.
- Formulated and solved an optimization problem to compensate for ultrasound image distortions induced by temperature gradients in weld bead inspections.
- Implemented a parametric model to estimate acoustic refraction profiles between heterogeneous media using ultrasonic data.
- Implemented and evaluated multiple ultrasonic image reconstruction algorithms for non-destructive testing applications.

Laboratory of Electronic Control of Electrical Machines

Curitiba, Brazil

UNDERGRADUATE RESEARCHER

Apr. 2019 - Mar. 2021

- **Advisor:** *Walter D. Sanchez*
- Implemented numerical methods in MATLAB for electrical engineering applications, including power flow analysis and electrical machine modeling.
- Programmed, deployed, and tested electrical machine controllers (e.g., soft starters and PLCs) for industrial applications using WEG equipment.

Peer-reviewed publications

- [1] Pires, G. P.[†], **Kalid, T. E.**[†], de A. Prado, T., Costa, V. L., Pereira, G. R., Passarin, T. A. R., & Pipa, D. R. (2025). *An acoustic lens for displacement-free sectorial inspection of pipes with ultrasonic phased arrays*. *NDT & E International*, 156, 103459. <https://doi.org/10.1016/j.ndteint.2025.103459>. ([†] denotes equal contribution).
- [2] **Kalid, T. E.**, Everton Trento Jr, Tatiana A. Prado, Gustavo P. Pires, Giovanni A. Guarneri, Thiago A. R. Passarin, & Daniel R. Pipa. (2023). *Virtual encoder: a two-dimension visual odometer for NDT*. *Research and Review Journal of Nondestructive Testing*, 1(1). <https://doi.org/10.58286/28119>
- [3] Muller, M. Y., de Almeida Prado, T., **Kalid, T. E.**, Passarin, T. A. R., & Pipa, D. R. (2023). *Ultrasonic sectorial inspection in the presence of temperature gradients*. *Research and Review Journal of Nondestructive Testing*, 1(1). <https://doi.org/10.58286/28122>

Conference Proceedings

- [1] **Kalid, T. E.**^{*}, Lazzaretti, A., Prado, T., Pires, G., Pipa, D., & Passarin, T. (2026). *Machine learning-driven flaw detection for ultrasonic pipe inspections with acoustic lens*. Proceedings of the 14th European Conference on Non-Destructive Testing 2026 (ECNDT 2026), June 15-19, 2026, in Verona, Italy. <https://www.ndt.net/?id=33475>. (^{*} denotes presenting author).
- [2] Z. Y. Wang, F., Prado, T., **Kalid**^{*}, T., Pires, G., Brante, G., Pipa, D., & Passarin, T. (2026). *Modeling and Removal of Electronic Crosstalk Effects in Ultrasound Signals via Deconvolution*. Proceedings of the 14th European Conference on Non-Destructive Testing 2026 (ECNDT 2026), June 15-19, 2026 in Verona, Italy. <https://www.ndt.net/?id=33474>. (^{*} denotes presenting author).
- [3] Costa, V., **Kalid, T.**, Dário, M.^{*}, Prado, T., Pires, G., Pipa, D., & Passarin, T. (2026). *Suppression of geometrical artifacts from acoustic lenses through low-rank approximation*. Proceedings of the 14th European Conference on Non-Destructive Testing 2026 (ECNDT 2026), June 15-19, 2026 in Verona, Italy. <https://www.ndt.net/?id=33469>. (^{*} denotes presenting author).
- [4] Guarneri, G., Pegorini, V., Pipa, D., **Kalid, T.**^{*}, Prado, T., Pires, G., & Passarin, T. (2026). *mini-auspex: a Python library for ultrasonic data import, post-processing and imaging*. Proceedings of the 14th European Conference on Non-Destructive Testing 2026 (ECNDT 2026), June 15-19, 2026 in Verona, Italy. <https://www.ndt.net/?id=33472>. (^{*} denotes presenting author).
- [5] Dário, M.^{*}, Caetano, F., Santin, D., Guerra, H., Dos Santos, H., **Kalid, T.**, Prado, T., Pires, G., & Passarin, T. (2026). *Robust virtual encoder: Using ultrasound data for compensation of depth variations in camera-based odometry*. Proceedings of the 14th European Conference on Non-Destructive Testing 2026 (ECNDT 2026), June 15-19, 2026 in Verona, Italy. <https://www.ndt.net/?id=33355>. (^{*} denotes presenting author).

Patents & Intellectual property

- [1] Guarneri, G. A., Prado, T. A., Pipa, D. R., Passarin, T. A. R., **Kalid, T. E.**, & Pires, G. P. (2025). *System and Method for High-Speed Inspection of Circular Sections* (Portuguese title: *Sistema e Método para Inspeção de Seções Circulares em Alta Velocidade*). Patent application BR1020250294680. Filed on December 30, 2025. Instituto Nacional da Propriedade Industrial (INPI), Brazil.

Fellowships & Prizes

2026	Inventor Award (Prêmio Inventor) , Awarded by Petrobras in recognition of contributions to patent applications deposited in 2025.	Rio de Janeiro, Brazil
2025	3rd Best Thesis , Awarded by the Regional Council of Engineering (CREA-PR) for prominent electrical engineering thesis.	Paraná, Brazil
2025	Best in Class Honor , Awarded to the highest-achieving student (1st/60); ranked 8th among more than 1,600 Electrical Engineering graduates in the program's history (Top 0.5%).	Paraná, Brazil
2025	Petrobras Graduate R&D Fellowship , Awarded by the Brazilian Petroleum Company (Petrobras) to support graduate research assistants at the R&D project AUSPEX (Advanced Ultrasound Signal Processing for Equipment inspeCtionS).	Paraná, Brazil
2024	Ranked 1st , Selected as the top-ranked candidate for a full-tuition scholarship supporting an academic exchange period.	Paraná, Brazil
2023	PRH-ANP Undergraduate Fellowship , Awarded by the Brazilian National Agency of Petroleum, Natural Gas and Biofuels (ANP) to students with high academic performance and research potential in the energy sector	Brazil
2021	Petrobras Undergraduate R&D Fellowship , Awarded by the Brazilian Petroleum Company (Petrobras) to support undergraduate researchers at the R&D project AUSPEX	Paraná, Brazil

Skills

Programming & Computing	Python, MATLAB, C/C++, CUDA, NumPy, SciPy, Matplotlib, PyTorch
Tools & Platforms	Git, GitHub, Linux, $\text{\LaTeX}$, CIVA (by EXTENDE), AutoCAD, SolidWorks
Ultrasound Acquisition Systems	Panther (by Eddyfi), Gekko (by Eddyfi), Omniscan (by Olympus), Pioonier (by TPAC)
Additional Skills	Academic writing, Public speaking, 3D modelling and printing
Languages	Portuguese (Native), English (professional proficiency/C1), Spanish (B2)